



# Department of Computer Science

## University of Management and Technology

### Week 3 Status Report

**Project Title:** Campus Network (CCP)

**Instructor:** Sir Fahad Irshad

**Prepared By:** Hafiz Ahmad Islam Bhatti (Project Manager)

**Date:** June 10, 2026.

#### Executive Summary:

Week 3 focused on resolving the emulator instability reported in Week 2, finalizing network connectivity, and fully deploying the Captive Portal trap and security ACLs.

#### Resolution of Week 2 Blockers:

In our previous report, we noted severe console freezing when utilizing the c3745 EtherSwitch. To ensure the project could move forward, the team made a strategic pivot to optimize resources. We replaced the heavy switch image with a basic-level GNS3 GUI switch, which completely eliminated the CPU overhead and stabilized the topology.

#### Key Accomplishments This Week:

- **Security Implementation:** Strict ACLs were successfully applied, isolating Guest traffic and dropping pings to the internal Student and Staff VLANs.
- **Captive Portal Trap:** Deployed a headless Ubuntu Server running dnsmasq to override external DNS, forcing all HTTP traffic on VLAN 99 to resolve to the portal IP (192.168.99.2).
- **Rate Limiting:** Successfully capped the Guest sub-interface at 5 Mbps.

#### Technical Challenges & Mitigations:

- **dhclient Failure:** The Ubuntu client VMs lacked the dhclient package, preventing dynamic IP assignment from the core router. **Mitigation:** The router DHCP pools remain fully programmed, but we manually assigned static IPs to the clients to keep testing on schedule.
- **GUI-less Authentication:** Because we used a headless server for the Guest VM to save RAM, there was no graphical browser to click "Accept" on the portal. **Mitigation:** We successfully utilized the Linux curl command (`curl -X POST -d "accept=1" [http://192.168.99.2](http://192.168.99.2))` to trigger the PHP portal authentication and restore internet access.